

Jammertest 2026

Transmission Plan

2026-09-14 to 2026-09-18

Generated: 2026-09-01 16:05

Monday — 2026-09-14

Time	Test Area 1	Test Area 2	Test Area 3
08:00			
09:00			
10:00			
11:00	11:00-13:00 - 0.0.1 Mandatory morning briefing Contact: Christian Skjetne (NPRA) Comment: MANDATORY	11:00-13:00 - 0.0.1 Mandatory morning briefing Contact: Christian Skjetne (NPRA) Comment: MANDATORY attendance at Test Area 1 (Site 1)	11:00-13:00 - 0.0.1 Mandatory morning briefing Contact: Christian Skjetne (NPRA) Comment: MANDATORY attendance at Test Area 1 (Site 1)
12:00	<i>Ongoing event. See above.</i>	<i>Ongoing event. See above.</i>	<i>Ongoing event. See above.</i>
13:00	13:00-14:00 - 0.1.1 Grace period Contact: Christian Skjetne (NPRA) Comment: Lunch	13:00-14:00 - 0.1.1 Grace period Contact: Øystein Karlsen (NKOM) Comment: Lunch @ Grunhvatn (Site 2)	13:00-14:00 - 0.1.1 Grace period Contact: Jahn Erik Røhme (NPRA) Comment: Lunch @ Stave (Site 3)
14:00	14:00-14:10 - 1.2.1 Jamming: CW: L1 Contact: Jørn Skorstad (Nkom) Comment: CW jamming at L1 from F8.1 Porcus Maior at Ramnan. Power: 50 W (47 dBm) 14:25-14:35 - 1.2.6 Jamming: CW: L1, L2, L5, G1, G2, E5b, E6, B1I Contact: Jørn Skorstad (Nkom) Comment: CW jamming at all GNSS bands from F8.1 Porcus Maior at Ramnan. Power: 50 W (47 dBm) 14:50-15:00 - 1.3.1 Jamming: sweep: L1, 100 kHz Contact: Jørn Skorstad (Nkom) Comment: Frequency sweep jamming at L1 from F8.1 Porcus Maior at Ramnan. For bandwidth of the specific band, see Testcatalog Table 5.25. Power: 50 W (47 dBm)	14:00-14:12 - 1.1.1 Jammer S1.1 Contact: Øystein Karlsen (NKOM) 14:16-14:28 - 1.1.4 Jammer S2.1 Contact: Øystein Karlsen (NKOM) 14:32-14:44 - 1.1.8 Jammer U1.1 Contact: Øystein Karlsen (NKOM) 14:48-15:00 - 1.1.12 Jammer H1.1 Contact: Øystein Karlsen (NKOM)	14:00-14:23 - 2.6.6 Motorcade moving with fixed spoofed position Contact: Jahn Erik Røhme (NPRA) Comment: This actual test will take 5 minutes extra than the other tests, because the motorcade will stay stationary the first five minutes. We should arrive at Crossroads 14.23 which allows us to have a 7 minute break instead of the regular 10 min. Power: 0.001 W (0 dBm) 14:30-14:50 - 2.6.5 Spoofing: Motorcade moving from Crossroads northbound to Stave, while the spoofed position moves southbound towards Nordmela Contact: Jahn Erik Røhme (NPRA) Power: 0.001 W (0 dBm)
15:00	15:15-15:25 - 1.3.12 Jamming: sweep: L1, L2, L5, G1, G2, E5b, E6, B1I, 100 kHz Contact: Jørn Skorstad (Nkom) Comment: Frequency sweep jamming of all GNSS bands from F8.1 Porcus Maior at Ramnan. For bandwidths of the specific bands, see Testcatalog Table 5.25. Power: 50 W (47 dBm) 15:40-15:50 - 1.4.5 Jamming: PRN: L1, Chiprate: 10 MHz Contact: Jørn Skorstad (Nkom) Comment: PRN jamming at L1 from F8.1 Porcus Maior at Ramnan. Power: 50 W (47 dBm)	15:04-15:16 - 1.1.13 Jammer H1.2 Contact: Øystein Karlsen (NKOM) 15:20-15:32 - 1.1.16 Jammer H3.1 Contact: Øystein Karlsen (NKOM) 15:36-15:48 - 1.1.18 Jammer H3.3 Contact: Øystein Karlsen (NKOM) 15:52-16:04 - 1.1.19 Jammer H4.1 Contact: Øystein Karlsen (NKOM)	15:00-15:20 - 2.6.2 Spoofing: Motorcade moving with fixed spoofed position Contact: Jahn Erik Røhme (NPRA) Power: 0.001 W (0 dBm) 15:30-15:50 - 2.6.7 Spoofing: Motorcade is driving from Crossroads to Stave, while the spoofed location is following the road at a higher speed 100km/h Contact: Jahn Erik Røhme (NPRA) Power: 0.001 W (0 dBm)
16:00	16:05-16:15 - 1.4.7 Jamming: PRN: L1, L2, L5, G1, G2, E5b, E6, B1I Contact: Jørn Skorstad (Nkom) Comment: PRN jamming at all GNSS bands from F8.1 Porcus Maior at Ramnan. Power: 50 W (47 dBm) 16:30-17:45 - 1.8.2 Jamming: PRN pyramid starting with L1 Contact: Jørn Skorstad (Nkom) Comment: PRN pyramid jamming, starting with L1 from F8.1 Porcus Maior (for chiprates, see Testcatalog Table 5.25), then adding bands sequentially to G1, L2, L5, L6, B1I, E5b, G2. The test then continues by removing bands one by one in reverse order, until ending up with only L1. Each step lasts 5 minutes with 3 minutes of jamming and a 2 minutes break between each pyramid step. In total, it will look like this: L1 L1, G1 L1, G1, L2 L1, G1, L2, L5 L1, G1, L2, L5, E6 L1, G1, L2, L5, E6, B1I L1, G1, L2, L5, E6, B1I, E5b L1, G1, L2, L5, E6, B1I, E5b, G2 L1, G1, L2, L5, E6, B1I, E5b L1, G1, L2, L5, E6, B1I L1, G1, L2, L5, E6 L1, G1, L2, L5, E6	<i>Ongoing event. See above.</i> 16:08-16:20 - 1.1.20 Jammer H6.1 Contact: Øystein Karlsen (NKOM) 16:24-16:36 - 1.1.21 Jammer H6.2 Contact: Øystein Karlsen (NKOM) 16:40-16:52 - 1.1.22 Jammer H6.3 Contact: Øystein Karlsen (NKOM) 16:56-17:08 - 1.1.23 Jammer H6.4 Contact: Øystein Karlsen (NKOM)	16:00-16:40 - 2.6.10 Motorcade driving while the power of the spoofer is ramping up Contact: Jahn Erik Røhme (NPRA) Comment: The route will run from Stave to the turning point at the Crossroads and then back to Stave. There will be no stop at the turning point, allowing the motorcade to remain in continuous movement throughout the test while the power level is increasing. The motorcade will drive at a speed of 35km/h. The estimated driving time between Stave – Crossroads is 17 minutes when driving at a speed of 35km/h. The test signal will be on air for 40-minutes without any stop. There will be no initial jamming during this test The starting and end point is Stave Power: 0.001 W (0 dBm) 16:50-17:10 - 2.6.4 Spoofing: Motorcade is driving while the spoofed position is driving in the terrain next to the road. Spoofing route will be route 8 (See appendix F) Contact: Jahn Erik Røhme (NPRA) Power: 0.001 W (0 dBm)

Monday — 2026-09-14

Time	Test Area 1	Test Area 2	Test Area 3
	L1, G1, L2, L5 L1, G1, L2 L1, G1 L1 Power: 50 W (47 dBm)		
17:00	Ongoing event. See above.	Ongoing event. See above. 17:12-17:24 - 1.1.26 Jammer H8.1 Contact: Øystein Karlsen (NKOM) 17:28-17:40 - 1.1.27 Jammer F6.1 Contact: Øystein Karlsen (NKOM) 17:44-17:56 - 1.1.29 Jammer H2.1 Contact: Øystein Karlsen (NKOM)	Ongoing event. See above. 17:20-17:40 - 2.6.11 Spoofing: Position jump and driving through a tunnel Contact: Jahn Erik Røhme (NPRA) Power: 0.001 W (0 dBm) 17:50-18:05 - 2.6.12 Spoofing: The motorcade is stationary at Stave. The time will jump forward by 1 year, 1 day, 1 hour and 15 minutes Contact: Jahn Erik Røhme (NPRA) Power: 0.001 W (0 dBm)
18:00	18:30-19:00 - 0.0.2 Mandatory afternoon (de)briefing Contact: Christian Skjetne (NPRA)	18:30-19:00 - 0.0.2 Mandatory afternoon (de)briefing Contact: Jahn Erik Røhme (NPRA) Comment: MANDATORY attendance at Stave (Site 3)	Ongoing event. See above. 18:30-19:00 - 0.0.2 Mandatory afternoon (de)briefing Contact: Jahn Erik Røhme (NPRA) Comment: MANDATORY attendance at Stave (Site 3)
19:00			
20:00			
21:00			

Tuesday — 2026-09-15

Time	Test Area 1	Test Area 2	Test Area 3
08:00	08:00-08:30 - 0.0.1 Mandatory morning briefing Contact: Christian Skjetne (NPRA) Comment: MANDATORY	08:00-08:30 - 0.0.1 Mandatory morning briefing Contact: Christian Skjetne (NPRA) Comment: MANDATORY	08:00-08:30 - 0.0.1 Mandatory morning briefing Contact: Jahn Erik Røhme (NPRA) Comment: MANDATORY attendance at Stave (Site 3)
09:00	09:00-09:23 - 1.6.10 Jamming: 7.9 µW (-21dBm) to 50 W (47dBm) with 2 dB increments PRN: L1, sweep: Iridium STL, step time: 20 s Contact: Jørn Skorstad (Nkom) Comment: PRN jamming, from F8.1 Porcus Maior (for chiprates, see Testcatalog Table 5.25), at L1 and frequency sweep jamming within the Iridium band (1626 - 1626.5 MHz) with a power ramp from 7.9 µW (-21dBm) to a maximum of 50 W (47dBm) and down to 7.9 µW (-21dBm) with 2 dB increments. Power level step time is 20 seconds. Power: 50 W (47 dBm) 09:15-09:49 - 3.3.1 Meaconing: RX1 with ramping power Contact: Jørn Skorstad (Nkom) Comment: Meaconing from F1.1 Porcellus using receiver RX1, with ramping power. Power is ramped up from 0.001 W to 10 W and then down again to 0.001 W in 5 dB steps, with each step lasting for 2 minutes. Power: 10 W (40 dBm) 09:40-10:03 - 1.6.9 Jamming 7.9 µW (-21dBm) to 50 W (47dBm) with 2 dB increments PRN: L1, L2, L5, G1, G2, E5b, E6, B1I, step time: 20 s Contact: Jørn Skorstad (Nkom) Comment: PRN jamming, from F8.1 Porcus Maior (for chiprates, see Testcatalog Table 5.25), with a power ramp from 7.9 µW (-21dBm) to a maximum of 50 W (47dBm) and down to 7.9 µW again, with 2 dB increments, within the test bands L1, L2, L5, G1, G2, E5b, E6, B1I. Power level step time is 20 seconds Power: 50 W (47 dBm)	09:00-09:55 - 1.25.4 Circular PRN jamming power ramp with 2 dB increments, 30 sec pr. steps Contact: Øystein Karlsen (NKOM)	09:00-09:20 - 1.11.12 Driving past a parked car with one commercially available military training jammer Contact: Jahn Erik Røhme (NPRA) Comment: Jammers available for testing: H1.1 Power: 0.1 W (20 dBm) 09:30-09:50 - 1.11.13 Driving past two parked cars, each with one commercially available military training jammer Contact: Jahn Erik Røhme (NPRA) Comment: Jammers available for testing: H1.1, H1.1 Power: 0.1 W (20 dBm)
10:00	<i>Ongoing event. See above.</i> 10:20-10:30 - 3.1.3 Meaconing: RX1 at 10 W Contact: Jørn Skorstad (Nkom) Comment: Meaconing from F1.1 Porcellus at approximately 40 dBm from F1.1 Porcellus at Ramnan of the L1 and L2 bands using receiver RX1. Power: 10 W (40 dBm) 10:45-11:00 - 3.1.7 Meaconing: RX1 at 10 W with initial jamming from the community house Contact: Jørn Skorstad (Nkom) Comment: Meaconing from F1.1 Porcellus at approximately 40 dBm will be conducted from F1.1 Porcellus for 10 minutes of the L1 and L2 bands using receiver RX1. This will be preceded by 5 minutes of PRN jamming at 30 dBm from Bleik (PRN L1, L2, L5, G1, G2, E5b, E6 and B1I). The meaconing will begin 4 minutes after the start of the jamming, resulting in a 1-minute overlap between the two activities. Power: 10 W (40 dBm)	10:15-11:10 - 1.25.2 Circular PRN jamming power ramp with 4 dB increments, 1min pr. steps Contact: Øystein Karlsen (NKOM)	10:00-10:20 - 1.11.14 Driving past three parked cars, each with a jammer inside Contact: Jahn Erik Røhme (NPRA) Comment: Jammers available for testing: H1.1, H1.1 and H4.1 Power: 0.1 W (20 dBm) 10:30-11:10 - 1.11.15 One commercially available military training jammer in the center of the motorcade while driving on route 2 Contact: Jahn Erik Røhme (NPRA) Power: 0.1 W (20 dBm)
11:00	11:15-11:49 - 3.3.1 Meaconing: RX1 with ramping power Contact: Jørn Skorstad (Nkom) Comment: Meaconing with F1.1 Porcellus using receiver RX1, with ramping power. Power is ramped up from 0.001 W to 10 W and then down again in to 0.001 W in 5 dB steps, with each step lasting for 2 minutes. Power: 10 W (40 dBm)	<i>Ongoing event. See above.</i> 11:15-11:45 - 1.25.5 Circular PRN jamming power ramp with 4 dB increments, 30 sec pr. steps Contact: Øystein Karlsen (NKOM)	<i>Ongoing event. See above.</i> 11:20-12:00 - 1.11.16 Two commercially available military training jammers in the motorcade while driving on route 2 Contact: Jahn Erik Røhme (NPRA) Power: 0.1 W (20 dBm)
12:00	12:00-13:00 - 0.1.1 Grace period Contact: Christian Skjetne (NPRA) Comment: Lunch	12:00-13:00 - 0.1.1 Grace period Contact: Øystein Karlsen (NKOM) Comment: Lunch @ Grunnavatn (Site 2)	12:00-13:00 - 0.1.1 Grace period Contact: Jahn Erik Røhme (NPRA) Comment: Lunch @ Stave (Site 3)
13:00	13:05-13:41 - 3.2.7 Meaconing: RX1 and RX2 at 10 W alternating with different switching frequencies Contact: Jørn Skorstad (Nkom) Comment: Meaconing from F1.1 Porcellus with increasing switching frequency between two antennas: RX1 and RX2. The switching frequency will double every 2 minutes. The first switching frequency will be 1 minute (between RX1 and RX2), then 30 sec, then 15 sec, etc. until the end of the test. Power: 10 W (40 dBm)	13:00-13:55 - 1.25.3 Circular PRN jamming power ramp with 4 dB increments and pulse duration 1ms, 10 % duty cycle Contact: Øystein Karlsen (NKOM)	13:00-13:40 - 1.11.17 Three jammers in the motorcade while driving on route 2 Contact: Jahn Erik Røhme (NPRA) Comment: Jammer-equipment will be decided and made available in the testlog afterwards. Power: 0.1 W (20 dBm) 13:50-14:30 - 1.11.17 Three jammers in the motorcade while driving on route 2 Contact: Jahn Erik Røhme (NPRA) Comment: Jammer-equipment will be decided and made available in the testlog afterwards. Power: 0.1 W (20 dBm)
14:00	14:00-14:36 - 3.2.11 Meaconing: RX1 and RX2 at 10 W alternating with different switching frequencies and with jamming from a different location	14:00-14:30 - 1.25.5 Circular PRN jamming power ramp with 4 dB increments, 30 sec pr. steps Contact: Øystein Karlsen (NKOM)	<i>Ongoing event. See above.</i> 14:40-15:30 - 1.11.17

Tuesday — 2026-09-15

Time	Test Area 1	Test Area 2	Test Area 3
	Contact: Jørn Skorstad (Nkom) Comment: Meaconing from F1.1 Porcellus with increasing switching frequency between two pick-up antennas: RX1 and RX2 with jamming from the community house. The switching frequency will double every 2 minutes, starting with 1 minute switching frequency, then 30 sec., etc. Meaconing on L1 and L2, jamming on L5, G1, G2, B1I, E5b, E6 from the community house Power: 10 W (40 dBm) 14:55-15:10 - 1.18.5 Jammer F8.1: CW signal drift: 1545 to 1620 MHz, 15 minutes sweep time Contact: Jørn Skorstad (Nkom) Comment: Out of band jamming from Porcus Maior at Ramnan Power: 50 W (47 dBm)	14:40-15:05 - 1.25.8 Circular AWGN jamming power ramp with 2 dB increments, 6 sec pr. step Contact: Øystein Karlsen (NKOM)	Three jammers in the motorcade while driving on route 2 Power: 0.1 W (20 dBm)
15:00	<i>Ongoing event. See above.</i> 15:25-15:40 - 1.18.7 Jammer F8.1: CW signal drift: 1620 to 1545 MHz, 15 minutes sweep time Contact: Jørn Skorstad (Nkom) Comment: Out of band jamming from Porcus Maior at Ramnan Power: 50 W (47 dBm) 15:55-16:10 - 1.18.13 Jammer F8.1: 50 W drift: 1150 to 1300 MHz, with CW and sweep time of 15 minutes Contact: Jørn Skorstad (Nkom) Comment: Out of band jamming from Porcus Maior at Ramnan Power: 50.1 W (47 dBm)	<i>Ongoing event. See above.</i> 15:10-16:05 - 1.25.6 Circular PRN jamming power ramp with 2 dB increments, 30 sec pr. steps from 1 SDR jammer and 3 CHIRP jammers Contact: Øystein Karlsen (NKOM)	<i>Ongoing event. See above.</i> 15:30-16:00 - 0.1.1 Grace period Contact: Jahn Erik Røhme (NPRA) Comment: Coffee break. We would appreciate receiving feedback on any ad hoc tests you would like to conduct from 16:50 onwards (see the test schedule). This will help us plan for the required personnel, vehicles and equipment, so we can try to accommodate your wishes.
16:00	<i>Ongoing event. See above.</i> 16:25-16:40 - 1.18.15 Jammer F8.1: 50 W drift: 1300 to 1150 MHz, with CW and sweep time of 15 minutes Contact: Jørn Skorstad (Nkom) Comment: Out of band jamming from Porcus Maior at Ramnan Power: 50 W (47 dBm)	<i>Ongoing event. See above.</i> 16:15-16:39 - 1.25.9 Circular AWGN jamming power ramp with 5 dB increments, 30 sec pr. step Contact: Øystein Karlsen (NKOM) 16:50-17:45 - 1.25.7 Circular PRN jamming power ramp with 2 dB increments, 30 sec pr. steps from 3 SDR jammer and 3 CHIRP jammers Contact: Øystein Karlsen (NKOM)	16:00-16:40 - 1.11.17 Three jammers in the motorcade while driving on route 2 Contact: Jahn Erik Røhme (NPRA) Power: 0.1 W (20 dBm) 16:50-18:00 - 0.3.1 Ad hoc test Contact: Jahn Erik Røhme (NPRA) Comment: What would you like to test? Please ask us and we will see if we can accommodate your wishes. If there are any previous tests you would like to repeat, or if you have suggestions for new tests that could provide valuable insights, please let us know. Spoofing scenarios are unfortunately not available. Power: 0.1 W (20 dBm)
17:00	17:00-17:28 - 3.4.5 Meaconing and spoofing from two emitters at two locations in sequence. Contact: Jørn Skorstad (Nkom) Comment: 5 minutes per emitter for 20 minutes, then 2 minutes each for the last 8 minutes. 10 second overlapping when switching emitter. Meaconing will be conducted with a power level of approximately 40 dBm from Porcellus at Ramnan, while spoofing will be at 25 dBm from the antenna at the community house. Power: 10 W (40 dBm) 17:45-18:00 - 3.4.1 Meaconing and spoofing at the same time from different locations. Contact: Jørn Skorstad (Nkom) Comment: Meaconing from Ramnan and spoofing from the community house at the same time. Meaconing will be conducted with a power level of approximately 40 dBm from Porcellus at Ramnan, while spoofing will be at 25 dBm from the antenna at the community house. Power: 10 W (40 dBm)	<i>Ongoing event. See above.</i>	<i>Ongoing event. See above.</i>
18:00	18:30-19:00 - 0.0.2 Mandatory afternoon (de)briefing Contact: Christian Skjetne (NPRA)	18:30-19:00 - 0.0.2 Mandatory afternoon (de)briefing Contact: Christian Skjetne (NPRA) Comment: MANDATORY attendance at Stave (Site 3)	18:30-19:00 - 0.0.2 Mandatory afternoon (de)briefing Contact: Jahn Erik Røhme (NPRA) Comment: MANDATORY attendance at Stave (Site 3)
19:00	19:00-22:00 - 1.16.7 High Power PRN jamming from two locations: L1, G1, B1I, L2, L5, E5b, E6, G2, Iridium Contact: Jørn Skorstad (Nkom) Comment: One hour of high power jamming from Porcus Maior at Ramnan, followed by one hour jamming from both Porcus Maior and the community house, followed by one hour of jamming from the community house. Periodic jamming in the Iridium band. Power output from the community house will be adjusted so that the field strength outside the community house is approximately equal to that from Porcus Maior. Power: 50 W (47 dBm)		
20:00	<i>Ongoing event. See above.</i>		
21:00	<i>Ongoing event. See above.</i>		

Wednesday — 2026-09-16

Time	Test Area 1	Test Area 2	Test Area 3
08:00	08:00-08:30 - 0.0.1 Mandatory morning briefing Contact: Christian Skjetne (NPRA) Comment: MANDATORY	08:00-08:30 - 0.0.1 Mandatory morning briefing Contact: Christian Skjetne (NPRA) Comment: MANDATORY attendance at Test Area 1 (Site 1)	08:00-08:30 - 0.0.1 Mandatory morning briefing Contact: Jahn Erik Røhme (NPRA) Comment: MANDATORY attendance at Stave (Site 3)
09:00	09:00-09:40 - 2.1.1 Large position and time jump, with power ramp Contact: Jørn Skorstad (Nkom) Comment: Signals: GPS L1 C/A, L2C, L5. Galileo E1, E5 E6. No jamming. Simulated position: 70 N, 10 E. Simulated start time: 01.10.2026 12:00. Power will be ramped up from -35 dBm to 25 dBm in 5 dB steps, with each step lasting 3 minutes. Power: 0.316 W (25 dBm)	09:00-09:40 - 2.10.1 Spoofing route GPS L1 and Galileo E1 only Contact: Øystein Karlsen (NKOM)	09:00-09:20 - 1.11.12 Driving past a parked car with one commercially available military training jammer Contact: Jahn Erik Røhme (NPRA) Comment: Jammers available for testing: H1.1 Power: 0.1 W (20 dBm) 09:30-09:50 - 1.11.13 Driving past two parked cars, each with one commercially available military training jammer Contact: Jahn Erik Røhme (NPRA) Comment: Jammers available for testing: H1.1, H1.1 Power: 0.1 W (20 dBm)
10:00	10:00-10:15 - 2.1.3 Large position and time jump. Galileo E1 only Contact: Jørn Skorstad (Nkom) Comment: Signals: Galileo E1. No jamming. Simulated position: 70 N, 10 E. Simulated start time: 01.10.2026 12:00. Power: 0.316 W (25 dBm) 10:20-10:35 - 2.1.2 Large position and time jump. GPS L1 C/A only Contact: Jørn Skorstad (Nkom) Comment: Signals: GPS L1 C/A. No jamming. Simulated position: 70 N, 10 E. Simulated start time: 01.10.2026 12:00. Power: 0.316 W (25 dBm) 10:40-10:55 - 2.1.5 Large position and time jump. GPS and Galileo. Contact: Jørn Skorstad (Nkom) Comment: Signals: GPS L1 C/A, L2C, L5. Galileo E1, E5a, E5b, E6. No jamming. Simulated position: 70 N, 10 E. Simulated start time: 01.10.2025 12:00. Power: 0.316 W (25 dBm)	10:00-10:55 - 2.10.2 Circle of 3 stationary jammers, L1, L2 and spoofing route GPS L1 and Galileo E1 only (Chirp jamming) Contact: Øystein Karlsen (NKOM)	10:00-10:20 - 1.11.14 Driving past three parked cars, each with a jammer inside Contact: Jahn Erik Røhme (NPRA) Comment: Jammers available for testing: H1.1, H1.1 and H4.1 Power: 0.1 W (20 dBm) 10:30-11:10 - 1.11.15 One commercially available military training jammer in the center of the motorcade while driving on route 2 Contact: Jahn Erik Røhme (NPRA) Comment: Jammers available for testing: H1.1 Power: 0.1 W (20 dBm)
11:00	11:25-11:40 - 2.1.15 Simulated driving (route 1). GPS L1 C/A and Galileo E1 Contact: Jørn Skorstad (Nkom) Comment: Signals: GPS L1 C/A. Galileo E1. 5 minutes of initial jamming (L1, G1, B1I, E6, L2, E5b, L5 with 2 W) prior to spoofing transmission. Simulated start position: Bleik community house parking lot. Simulated start time: 01.10.2026 12:00. Power: 0.316 W (25 dBm) 11:45-12:00 - 2.1.10 Simulated driving (route 1), with initial jamming Contact: Jørn Skorstad (Nkom) Comment: Signals: GPS L1 C/A, L2C, L5. Galileo E1, E5, E6. 5 minutes of initial jamming (L1, G1, B1I, E6, L2, E5b, L5 with 2 W) prior to spoofing transmission. Simulated start position: Bleik community house parking lot. Simulated start time: 01.10.2026 12:00. Power: 0.316 W (25 dBm)	11:00-11:55 - 2.10.5 Circle of 3 stationary jammers, L1, L2 and spoofing route GPS L1 and Galileo E1 only (Narrow band jamming) Contact: Øystein Karlsen (NKOM)	<i>Ongoing event. See above.</i> 11:20-12:00 - 1.11.16 Two commercially available military training jammers in the motorcade while driving on route 2 Contact: Jahn Erik Røhme (NPRA) Power: 0.1 W (20 dBm)
12:00	12:00-13:00 - 0.1.1 Grace period Contact: Christian Skjetne (NPRA) Comment: Lunch	12:00-13:00 - 0.1.1 Grace period Contact: Øystein Karlsen (NKOM) Comment: Lunch @ Grunnvatn (Site 2)	12:00-13:00 - 0.1.1 Grace period Contact: Jahn Erik Røhme (NPRA) Comment: Lunch @ Stave (Site 3)
13:00	13:00-13:15 - 2.8.1 EGNOS with "Do Not Use GPS" commands Contact: Jørn Skorstad (Nkom) Comment: Signals: EGNOS L1. No jamming. The transmission of false SBAS signals will start immediately upon test start. Power: 1 W (30 dBm) 13:30-13:45 - 2.8.2 EGNOS with "Do Not Use GPS" commands and normali spoofing Contact: Jørn Skorstad (Nkom) Comment: Signals: EGNOS L1. No jamming. The transmission of false SBAS signals will start immediately upon test start. Test will also include coherent spoofing of GPS L1 C/A, L2, L5, E1 and E5 to ensure compliance between SBAS corrections and the GNSS signals. The spoofed position will be in front of the HQ. Power: 1 W (30 dBm)	13:05-13:45 - 2.10.6 Circle of 3 stationary SDR jammers, L1, L2 and spoofing route GPS L1 and Galileo E1 only (power ramping) Contact: Øystein Karlsen (NKOM)	13:00-13:40 - 1.11.17 Three jammers in the motorcade while driving on route 2 Contact: Jahn Erik Røhme (NPRA) Comment: Jammer-equipment will be decided and made available in the testlog afterwards. Power: 0.1 W (20 dBm) 13:50-14:30 - 1.11.17 Three jammers in the motorcade while driving on route 2 Contact: Jahn Erik Røhme (NPRA) Comment: Jammer-equipment will be decided and made available in the testlog afterwards. Power: 0.1 W (20 dBm)
14:00	14:00-14:15 - 0.3.1 Ad hoc test	14:00-14:40 - 2.10.1 Spoofing route GPS L1 and Galileo E1 only	<i>Ongoing event. See above.</i>

Wednesday — 2026-09-16

Time	Test Area 1	Test Area 2	Test Area 3
	Contact: Jørn Skorstad (Nkom) Comment: EGNOS test, details TBD Power: 0.316 W (25 dBm) 14:45-15:25 - 2.2.9 Small position jump, with power ramp Contact: Jørn Skorstad (Nkom) Comment: Signals: GPS L1 C/A, L2C, L5. Galileo E1, E5, E6. Power will be ramped up from -35 dBm to 25 dBm in 5 dB steps, with each step lasting 3 minutes. Simulated position: North end of the football field - 69.27701401, 15.96932835, 45 m hae. Simulated start time: Referenced to live GPS-signals. Power: 0.316 W (25 dBm)	Contact: Øystein Karlsen (NKOM) Comment: Spoofing route that start at LOK2-ORIG, and goes out forming a spoofing circle above position A150, B150 and C150. Spoofing signal will perform power ramp from uW to mW during the first 30 minutes of the test. Signals: GPS L1 C/A. Galileo E1. No initial jamming. Spoofing route duration is 40 minutes. Power: 0.001 W (0 dBm)	14:40-15:30 - 1.11.17 Three jammers in the motorcade while driving on route 2 Contact: Jahn Erik Røhme (NPRA) Comment: Jammer-equipment will be decided and made available in the testlog afterwards. Power: 0.1 W (20 dBm)
15:00	<i>Ongoing event. See above.</i> 15:30-15:45 - 2.2.2 Small position jump, with initial and continuous jamming Contact: Jørn Skorstad (Nkom) Comment: Signals: GPS L1 C/A, L2C, L5. Galileo E1, E5, E6. 5 minutes of initial jamming. Power: 0.316 W (25 dBm)	15:00-15:55 - 2.10.2 Circle of 3 stationary jammers, L1, L2 and spoofing route GPS L1 and Galileo E1 only (Chirp jamming) Contact: Øystein Karlsen (NKOM)	<i>Ongoing event. See above.</i> 15:30-16:00 - 0.1.1 Grace period
16:00	16:15-16:25 - 2.3.5 Simulated driving (route 1). GPS only Contact: Jørn Skorstad (Nkom) Comment: Signals: GPS L1 C/A, L2C, L5. No jamming. Simulated start position: Bleik community house parking lot. Simulated start time: Referenced to live GPS-signals. Power: 0.316 W (25 dBm) 16:30-16:40 - 2.3.10 Simulated driving (route 1) Contact: Jørn Skorstad (Nkom) Comment: Signals: GPS L1 C/A, L2C, L5. Galileo E1, E5, E6. No jamming. Simulated start position: Bleik community house parking lot. Simulated start time: Referenced to live GPS-signals. Power: 0.316 W (25 dBm) 16:45-17:00 - 2.3.11 Simulated driving (route 1) with initial and continuous jamming. Contact: Jørn Skorstad (Nkom) Comment: Signals: GPS L1 C/A, L2C, L5. Galileo E1, E5, E6. 5 minutes of initial jamming (L1, G1, B1I, E6, L2, E5b, L5 with 2 W) prior to spoofing transmission, then continuous on other bands than the ones spoofed. Simulated start position: Bleik community house parking lot. Simulated start time: Referenced to live GPS-signals. Power: 0.316 W (25 dBm)	16:00-16:55 - 2.10.5 Circle of 3 stationary jammers, L1, L2 and spoofing route GPS L1 and Galileo E1 only (Narrow band jamming) Contact: Øystein Karlsen (NKOM)	16:00-16:40 - 1.11.17 Three jammers in the motorcade while driving on route 2 Contact: Jahn Erik Røhme (NPRA) Comment: Jammer-equipment will be decided and made available in the testlog afterwards. Power: 0.1 W (20 dBm) 16:50-18:00 - 0.3.1 Ad hoc test Contact: Jahn Erik Røhme (NPRA) Comment: What would you like to test? Please ask us and we will see if we can accommodate your wishes. If there are any previous tests you would like to repeat, or if you have suggestions for new tests that could provide valuable insights, please let us know. Spoofing scenarios are unfortunately not available. Power: 0.1 W (20 dBm)
17:00	17:15-17:30 - 2.3.18 GPS PRN 37 receiver robustness test scenario. GPS L1 only Contact: Jørn Skorstad (Nkom) Comment: Signals: GPS L1 C/A. No jamming. Fixed spoofed position: 69.27547832, 15.96832496, 35 m hae. Scenario: One satellite is configured as PRN 37 from the start of the simulation and remains PRN 37 throughout the scenario. This tests receiver behaviour when a reserved PRN is present during acquisition and tracking. The simulated PRN 37 signal will be transmitted with satellite health set to healthy. Power: 0.316 W (25 dBm) 17:45-18:00 - 2.3.13 Flying (route 4) - "drone scenario" Contact: Jørn Skorstad (Nkom) Comment: Drone "Jammertest" scenario. Initially 5 min static position. followed by 10 min of movement. Power: 0.316 W (25 dBm)	17:05-17:45 - 2.10.6 Circle of 3 stationary SDR jammers, L1, L2 and spoofing route GPS L1 and Galileo E1 only (power ramping) Contact: Øystein Karlsen (NKOM)	<i>Ongoing event. See above.</i>
18:00	18:30-19:00 - 0.0.2 Mandatory afternoon (de)briefing Contact: Christian Skjetne (NPRA) Comment: MANDATORY	18:30-19:00 - 0.0.2 Mandatory afternoon (de)briefing Contact: Jahn Erik Røhme (NPRA) Comment: MANDATORY attendance at Stave (Site 3)	18:30-19:00 - 0.0.2 Mandatory afternoon (de)briefing Contact: Jahn Erik Røhme (NPRA) Comment: MANDATORY attendance at Stave (Site 3)
19:00	19:00-22:00 - 2.3.19 Long period with circular position drift Contact: Jørn Skorstad (Nkom) Comment: First hour: 5 min static, then drifting 0.1 m/s Second hour: 5 min static, then drifting 3 m/s Third hour: 5 min static, then drifting 30 m/s Power: 0.316 W (25 dBm)		
20:00	<i>Ongoing event. See above.</i>		

Wednesday — 2026-09-16			
Time	Test Area 1	Test Area 2	Test Area 3
21:00	Ongoing event. See above.		

Thursday — 2026-09-17

Time	Test Area 1	Test Area 2	Test Area 3
08:00	08:00-08:30 - 0.0.1 Mandatory morning briefing Contact: Christian Skjetne (NPRA) Comment: MANDATORY	08:00-08:30 - 0.0.1 Mandatory morning briefing Contact: Christian Skjetne (NPRA) Comment: MANDATORY attendance at Test Area 1 (Site 1)	08:00-08:30 - 0.0.1 Mandatory morning briefing Contact: Jahn Erik Røhme (NPRA) Comment: MANDATORY attendance at Stave (Site 3)
09:00	09:00-09:40 - 2.5.2 Time offset 15 minutes from real time, with power ramp Contact: Jørn Skorstad (Nkom) Comment: Signals: GPS L1 C/A, L2C, L5. Galileo E1, E5, E6. Fixed spoofed position: 69.27547832, 15.96832496, 35 m hae. Time offset is + 15 minutes (900 seconds), so "into the future".	09:00-09:55 - 3.5.1 Power ramp meaoning test 2 dB increments, 30 sec pr. steps Contact: Øystein Karlsen (NKOM)	09:00-09:23 - 2.6.6 Spoofing: Motorcade moving with fixed spoofed position Contact: Jahn Erik Røhme (NPRA) Comment: NB! This actual test will take 5 minutes extra than the other tests, because the motorcade will stay stationary the first five minutes. We should arrive at Crossroads 13.23 which allows us to have a 7 minute break instead of the regular 10 min. Power: 0.001 W (0 dBm) 09:30-09:50 - 2.6.5 Spoofing: Motorcade moving from Crossroads northbound to Stave, while the spoofed position moves southbound towards Nordmela Contact: Jahn Erik Røhme (NPRA) Power: 0.001 W (0 dBm)
10:00	10:00-10:40 - 2.4.14 Time offset -3 minutes from real time, with power ramp Contact: Jørn Skorstad (Nkom) Comment: Signals: GPS L1 C/A, L2C, L5. Galileo E1, E5, E6. Fixed spoofed position: 69.27547832, 15.96832496, 35 m hae. Time offset is - 3 minutes (180 seconds), so "back into the past". Power: 0.001 W (0 dBm) 10:55-11:30 - 2.4.13 Static + Pseudorange error, with initial and continous jamming Contact: Jørn Skorstad (Nkom) Comment: Signals: GPS L1 C/A, L2C, L5. Galileo E1, E5, E6. Fixed spoofed position: 69.27547832, 15.96832496, 35 m hae. 5 minutes of initial jamming (L1, G1, B1I, L2, E5b, L5 with 2 W) prior to spoofing transmission, then continuous on other bands than the ones spoofed. The pseudorange error is applied to all satellites, starting five minutes after the spoofing starts. The applied error should equal a drift of 3 ns/s. Power: 0.001 W (0 dBm)	10:00-10:55 - 3.5.6 Power ramp meaoning test 2 dB increments, 30 sec pr. steps and 1 SDR jammer PRN Contact: Øystein Karlsen (NKOM)	10:00-10:20 - 2.6.2 Spoofing: Motorcade moving with fixed spoofed position Contact: Jahn Erik Røhme (NPRA) Power: 0.001 W (0 dBm) 10:30-10:50 - 2.6.7 Spoofing: Motorcade is driving from Crossroads to Stave, while the spoofed location is following the road at a higher speed 100km/h Contact: Jahn Erik Røhme (NPRA) Power: 0.001 W (0 dBm)
11:00	<i>Ongoing event. See above.</i> 11:45-12:00 - 2.4.15 UNIX time rollover test 19.01.2038 (incoherent) Contact: Jørn Skorstad (Nkom) Comment: UNIX time rollover test. Starting time 03:04:00 19.01.2038 UTC. Power: 0.001 W (0 dBm)	11:00-11:55 - 3.5.7 Power ramp meaoning test 2 dB increments, 30 sec pr. steps + 3 SDR jammers Contact: Øystein Karlsen (NKOM)	11:00-11:40 - 2.6.10 Motorcade driving while the power of the spoofer is ramping up Contact: Jahn Erik Røhme (NPRA) Comment: The route will run from Stave to the turning point at the Crossroads and then back to Stave. There will be no stop at the turning point, allowing the motorcade to remain in continuous movement throughout the test while the power level is increasing. The motorcade will drive at a speed of 35km/h. The estimated driving time between Stave – Crossroads is 17 minutes when driving at a speed of 35km/h. The test signal will be on air for 40-minutes without any stop. There will be no initial jamming during this test The starting and end point is Stave Power: 0.001 W (0 dBm)
12:00	12:00-13:00 - 0.1.1 Grace period Contact: Christian Skjetne (NPRA) Comment: Lunch	12:00-13:00 - 0.1.1 Grace period Contact: Øystein Karlsen (NKOM) Comment: Lunch @ Grunnavatn (Site 2)	12:00-13:00 - 0.1.1 Grace period Contact: Jahn Erik Røhme (NPRA) Comment: Lunch @ Stave (Site 3)
13:00	13:00-13:35 - 2.5.25 Static + UTC-parameter nav. data manipulation (adding leap seconds), with initial and continuous jamming Contact: Jørn Skorstad (Nkom) Comment: Signals: GPS L1 C/A, L2C, L5. Galileo E1, E5, E6. 5 minutes of initial jamming (L1, G1, B1I, L2, E5b, L5 with 2 W) prior to spoofing transmission, then continuous on other bands than the ones spoofed. Fixed spoofed position: 69.27547832, 15.96832496, 35 m hae. The UTC-paramter nav. data manipulation consists of the spoofing signal saying that back in 2016, there was 19 leap seconds instead of 18. Power: 0.001 W (0 dBm)	13:00-13:55 - 3.5.1 Power ramp meaoning test 2 dB increments, 30 sec pr. steps Contact: Øystein Karlsen (NKOM)	13:00-13:23 - 2.6.6 Spoofing: Motorcade moving with fixed spoofed position Contact: Jahn Erik Røhme (NPRA) Comment: NB! This actual test will take 5 minutes extra than the other tests, because the motorcade will stay stationary the first five minutes. We should arrive at Crossroads 13.23 which allows us to have a 7 minute break instead of the regular 10 min. Power: 0.001 W (0 dBm) 13:30-13:50 - 2.6.5 Spoofing: Motorcade moving from Crossroads northbound to Stave, while the spoofed position moves southbound towards Nordmela Contact: Jahn Erik Røhme (NPRA) Power: 0.001 W (0 dBm)
14:00	14:00-14:15 - 2.5.5 Time offset 15 minutes from real time. Galileo E1 Contact: Jørn Skorstad (Nkom) Comment: Signals: Galileo E1. No jamming. Fixed spoofed position: 69.27547832, 15.96832496, 35 m hae. Time offset is + 15 minutes (900 seconds), so "into the future". Power: 0.001 W (0 dBm)	14:00-14:50 - 3.5.5 Pulsed meaoning and jamming (1 SDR jammer) Contact: Øystein Karlsen (NKOM)	14:00-14:20 - 2.6.2 Spoofing: Motorcade moving with fixed spoofed position Contact: Jahn Erik Røhme (NPRA) Power: 0.001 W (0 dBm) 14:30-14:50 - 2.6.7 Spoofing: Motorcade is driving from Crossroads to Stave, while the spoofed location is following the road at a higher speed 100km/h

Thursday — 2026-09-17

Time	Test Area 1	Test Area 2	Test Area 3
	14:20-14:35 - 2.5.4 Time offset 15 minutes from real time. GPS L1 C/A Contact: Jørn Skorstad (Nkom) Comment: Signals: GPS L1 C/A. No jamming. Fixed spoofed position: 69.27547832, 15.96832496, 35 m hae. Time offset is + 15 minutes (900 seconds), so "into the future". Power: 0.001 W (0 dBm) 14:40-14:55 - 2.5.6 Time offset 15 minutes from real time Contact: Jørn Skorstad (Nkom) Comment: Signals: GPS L1 C/A, L2C, L5. Galileo E1, E5, E6. No jamming. Fixed spoofed position: 69.27547832, 15.96832496, 35 m hae. Time offset is + 15 minutes (900 seconds), so "into the future". Power: 0.001 W (0 dBm)		Contact: Jahn Erik Røhme (NPRA) Power: 0.001 W (0 dBm)
15:00	15:20-16:00 - 2.5.15 Static + Pseudorange error Contact: Jørn Skorstad (Nkom) Comment: Signals: GPS L1 C/A, L2C, L5. Galileo E1, E5, E6. No jamming. Fixed spoofed position: 69.27547832, 15.96832496, 35 m hae. The pseudorange error is applied to all satellites, starting five minutes after the spoofing starts. The applied error is gradually increased throughout the test. Power: 0.001 W (0 dBm)	15:00-15:55 - 3.5.3 Power ramp meaoning test 2 dB increments, 30 sec pr. steps, with 3 background jammers CHIRP Contact: Øystein Karlsen (NKOM)	15:00-15:40 - 2.6.10 Spoofing: Motorcade driving while the power of the spoofer is ramping up Contact: Jahn Erik Røhme (NPRA) Power: 0.001 W (0 dBm) 15:40-16:10 - 0.1.1 Grace period Comment: Coffee-break at Stave.
16:00	16:15-16:30 - 2.7.17 UNIX time rollover test 19.01.2038 (coherent) Contact: Jørn Skorstad (Nkom) Comment: UNIX time rollover, starting time 03:04:00 19.01.2038. Coherent. Power: 0.001 W (0 dBm) 16:55-17:20 - 2.5.26 Static + UTC-parameter nav. data manipulation (removing leap seconds). GPS L1 C/A Contact: Jørn Skorstad (Nkom) Comment: Signals: GPS L1 C/A. Fixed spoofed position: 69.27547832, 15.96832496, 35 m hae. The UTC-paramter nav. data manipulation consists of the spoofing signal saying that back in 2016, there was counter-factual extra amount of -127 leap seconds, which in total means that there is removed -145 leap seconds. Power: 0.001 W (0 dBm)	16:00-16:55 - 3.5.8 Power ramp meaoning test 2 dB increments, 30s pr. steps, with 3 background jammers CHIRP and 1 SDR jammer PRN Contact: Øystein Karlsen (NKOM)	<i>Ongoing event. See above.</i> 16:10-16:30 - 2.6.4 Spoofing: Motorcade is driving while the spoofed position is driving in the terrain next to the road. Spoofing route will be route 8 (See appendix F) Contact: Jahn Erik Røhme (NPRA) Power: 0.001 W (0 dBm) 16:40-17:00 - 2.6.11 Spoofing: Position jump and driving through a tunnel Contact: Jahn Erik Røhme (NPRA) Power: 0.001 W (0 dBm)
17:00	<i>Ongoing event. See above.</i> 17:30-18:00 - 2.5.28 Static + UTC-parameter nav. data manipulation (removing leap seconds), with initial and continuous jamming Contact: Jørn Skorstad (Nkom) Comment: 2.5.28 or 2.5.27 (with or without initial jamming, TBD) Power: 0.001 W (0 dBm)	17:00-17:55 - 3.5.9 Power ramp meaoning test 2 dB increments, 30s pr. steps, with 3 background jammers CHIRP and 3 SDR Jammers PRN Contact: Øystein Karlsen (NKOM)	17:00-17:30 - 0.1.1 Grace period Contact: Jahn Erik Røhme (NPRA) Comment: Used for preparations for test 2.6.9 17:30-17:45 - 2.6.9 Spoofing: The motorcade is stationary at Stave. The clock will jump forward in time by 12 years (to the year 2038) Contact: Jahn Erik Røhme (NPRA) Comment: Warning: This test could 'brick' some equipment. Power: 0.001 W (0 dBm)
18:00	18:30-19:00 - 0.0.2 Mandatory afternoon (de)briefing Contact: Christian Skjetne (NPRA) Comment: MANDATORY	18:30-19:00 - 0.0.2 Mandatory afternoon (de)briefing Contact: Christian Skjetne (NPRA) Comment: MANDATORY attendance at Bleik (Site 1)	18:30-19:00 - 0.0.2 Mandatory afternoon (de)briefing Contact: Christian Skjetne (NPRA) Comment: MANDATORY attendance at Bleik (Site 1)
19:00			
20:00			
21:00			

Friday — 2026-09-18

Time	Test Area 1	Test Area 2	Test Area 3
08:00	08:00-08:30 - 0.0.1 Mandatory morning briefing Contact: Christian Skjetne (NPRA) Comment: MANDATORY		
09:00	09:00-09:15 - 0.3.1 Ad hoc test Contact: Jørn Skorstad (Nkom) Comment: TBD 09:30-09:45 - 0.3.1 Ad hoc test Contact: Jørn Skorstad (Nkom) Comment: TBD	09:00-16:00 - 0.1.1 Grace period Contact: Øystein Karlsen (NKOM) Comment: SITE IS CLOSED!	09:00-16:00 - 0.1.1 Grace period Contact: Jahn Erik Røhme (NPRA) Comment: SITE IS CLOSED!
10:00	10:00-11:00 - 1.24.37 The symphony Contact: Jørn Skorstad (Nkom) Comment: Sequential powering on of jammers with two minute intervals: 5 small jammers at Bleik (one lifted), Ramnan, spoofing from cemetery, and finally meaconing from the community house. After 14 minutes all jammers are turned on, and will stay on for the rest of the hour. Small jammers: L1. High power jammer: L1, L2, L5, G1, G2, E5b, E6, B1I. Meaconing on all bands.	Ongoing event. See above.	Ongoing event. See above.
11:00	11:15-11:35 - 0.3.1 Ad hoc test Contact: Jørn Skorstad (Nkom) Comment: TBD	Ongoing event. See above.	Ongoing event. See above.
12:00	12:00-13:00 - 0.1.1 Grace period Contact: Christian Skjetne (NPRA) Comment: Lunch	Ongoing event. See above.	Ongoing event. See above.
13:00	13:00-13:35 - 0.3.1 Ad hoc test Contact: Jørn Skorstad (Nkom) Comment: TBD 13:45-14:00 - 0.3.1 Ad hoc test Contact: Jørn Skorstad (Nkom) Comment: TBD	Ongoing event. See above.	Ongoing event. See above.
14:00	14:00-14:30 - 0.0.2 Mandatory afternoon (de)briefing Contact: Christian Skjetne (NPRA) Comment: Summary briefing and closing of event	Ongoing event. See above.	Ongoing event. See above.
15:00		Ongoing event. See above.	Ongoing event. See above.
16:00			
17:00			
18:00			
19:00			
20:00			
21:00			